# The rule, the room, and which settings give a birth: how a curled torus's change into a flat one depends on the coefficient of the local term

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In a family of graph energies around combinatorial quantum gravity (CQG), a coefficient $\lambda$ multiplies the price of a surplus square. Only $\lambda = 1$ is CQG; everything here is at $\lambda > 1$, in our family. There a torus with one direction curled lies $4(\lambda - 1)$ per vertex above flat and gives way after an activated first exit. We map each ingredient of that change against $\lambda$. Exact: the wall, the cheapest single move out, costs $32 - 16\lambda$ below $\lambda = 4/3$ and $64 - 40\lambda$ above, and counting the exits offered gives a first-exit time of 8330 to 183 sweeps for $\lambda = 1.05$ to 1.35 at coupling $g = 1.5$. Measured: first exits counted by the move are on that count at $\lambda = 1.05$ and 1.30. Three registered maps of the conversion were inconclusive; a fourth, timed without the earlier threshold detector, met its criteria in every cell from $\lambda = 1.05$ to 1.30 at 64 to 192 vertices, and at a held-out size of 288 predicted from those. Read from their wiring, 9,858 of 9,865 flat end states have the least width a flat torus can have, six edges around, and those sampled are the tube re-glued two steps along: the clean product does not widen. Sealed, the reservoir a clean conversion needs grows at least in proportion to the release, at one size. With six and eight links no setting opened a curled state into one flat space by its registered criterion; the published six-link curve has not been reproduced. A larger $\lambda$ gives a larger release, a shorter-lived curled state and a harder birth.

## I. INTRODUCTION

In a family of graph energies around combinatorial quantum gravity (CQG) [4–6], a coefficient $\lambda$ multiplies the price of a surplus square, and only $\lambda = 1$ is CQG. The companion paper [1] studied a tube at $\lambda > 1$: a $4 \times L$ torus with one direction curled to a period of four edges. The tube is stable for now: no single move out of it lowers its energy, yet lower states exist. After an activated first exit it converts toward a flat torus and releases energy, a change that paper calls a burp. Most of that work is at one setting, $\lambda = 1.25$, and one setting could be a lucky point. This paper asks how each ingredient of the change depends on $\lambda$: the wall (the cost of the cheapest single move out), the wait, the release, the room the product needs when the system is sealed, and the scrap left behind. It also summarizes what the same price list gives with more directions, which a companion paper treats in full [3].

*Why a coefficient at all.* At $\lambda = 1$ an edge carrying three squares costs what an edge carrying two costs, so the flat torus, the tube and the 4-cube have the same energy and a change between them releases nothing. For $\lambda > 1$ the zero-energy states are the graphs with two squares on every edge, and each curled direction costs a known energy. The states, the moves and the hard-core rule are those of CQG; the energy at $\lambda \neq 1$ is not its curvature action. We call the $\lambda \neq 1$ energies our family, or the toy. We have not found published work that sets the surplus price to any finite value above its curvature value: in print, as far as we have found, are $\lambda = 0$, $\lambda = 1$ and, in the text of two papers [5, 7], a version in which surplus squares are forbidden outright [8].

*Four kinds of statement* are kept apart and labeled. *Exact*: arithmetic or complete enumeration. *Measured*: a run registered before it ran, quoted with its registered verdict in capitals and with whose prediction held or failed. *Interpretation*: ours, unverified. *The author's*: the hypothesis that motivates the work. In that hypothesis the extra term is a curling cost whose value is whatever reality has, and the setting observed is the one whose cycle yields the most observers who evolved; that framing is not a claim of this paper. Readings made after a result was known are labeled retrospective. Test numbers such as T60 refer to registrations in the project's record [13]. Every kinetic statement is about one Markov chain, the local edge switch with Metropolis acceptance [1]; we do not claim it is a physical time evolution.

## II. THE PRICE LIST: WHAT IS EXACT

A state is a simple four-regular graph on a fixed labeled bipartition of $N$ vertices obeying the hard-core rule: no two vertices share more than two neighbors [5, 6]. With $S$ the number of squares (four-cycles), $S_e$ the number on edge $e$ and $X = \sum_e (S_e - 2)_+$ the surplus squares,

$$\begin{aligned} H &= 16\,(N - S) + 4\lambda X \\ &= 4 \sum_e \big[(2 - S_e) + \lambda\,(S_e - 2)_+\big], \end{aligned} \tag{1}$$

the second line using $\sum_e S_e = 4S$ over the $2N$ edges. A missing square costs 16 and a surplus square $4\lambda$. Since $2 - S_e = (2 - S_e)_+ - (S_e - 2)_+$, each edge contributes $4[(2 - S_e)_+ + (\lambda - 1)(S_e - 2)_+] \geq 0$ for $\lambda \geq 1$: no state lies below zero, and for $\lambda > 1$, $H = 0$ if and only if every edge carries exactly two squares [1].

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*The ladder.* Curling one direction of a torus to period four makes each ring of four a square, a third square on each of the $N$ edges around the curl; the $N$ edges along it keep two. So the tube (even $L > 4$) has $S = 5N/4$, $X = N$ and lies $\varepsilon N$ above zero, with $\varepsilon = 4(\lambda - 1)$; the 4-cube, both directions curled, lies $2\varepsilon$ per vertex above. A conversion that ends with graph energy $H_f$ releases $Q = \varepsilon N - H_f$, the full gap only when $H_f = 0$.

*The wall and the count.* Every switch the chain can propose out of the tube was listed and priced. Two kinds are cheapest: move A ($\Delta S = -2$, $\Delta X = -4$), costing $32 - 16\lambda$, in $3N$ ways; and move B ($\Delta S = -4$, $\Delta X = -10$), costing $64 - 40\lambda$, in $2N$ ways. The cheaper of the two is the wall: A below $\lambda = 4/3$, B above, both positive up to $\lambda = 1.6$. A proposal is an ordered pair of edges, $4N^2$ in all, and each switch arises from two of them; a sweep is $2N$ attempts. So a sweep offers A three times and B twice at every size. Weighting each offer by its acceptance gives the two-channel first-exit time at coupling $g$, the count:

$$\tau_{AB}(\lambda, g) = \left[ 3\, e^{-(32-16\lambda)/g} + 2\, e^{-(64-40\lambda)/g} \right]^{-1} \qquad (2)$$

(Table I). Dearer exits add at most 2% to the rate at the $g = 1.5$ conditions of the maps [1]. The count is the time to the first exit, not to the end of the conversion, and the wall is a first-exit cost, not a proven saddle of a complete conversion path. That $\tau_{AB}$ does not depend on $N$ is a property of the proposal, which draws its two edges from the whole graph. Durations in the maps below are in these chain sweeps, as in the earlier maps, because the fourth map repeats them. Where local rates are compared across sizes (the cooling times of Sec. VII) we use a size-fair clock: one size-fair sweep is $N/96$ chain sweeps, giving every local pair of edges the chances it has at 96 vertices. It changes how times compare across sizes, nothing else.

*The scrap's share.* The commonest relic of the cold sealed boxes of Ref. [1] is one column of the tube left curled: a closed loop of four vertices, costing $24\lambda - 16$ above flat. It does not vanish as $\lambda \to 1$, while the gap does. Its share of the gap is $(24\lambda - 16)/[4(\lambda - 1)N]$: 22% at $\lambda = 1.25$ and 72% at 1.05 for $N = 64$ (Table I), 7% and 24% for $N = 192$. Setting $24\lambda - 16 = 4(\lambda - 1)N$ gives $\lambda = 1 + 8/(4N - 24)$, which is 1.035 at $N = 64$; below it one such column costs more than the whole gap.

## III. THE WAIT

*Measured, counted by the move.* T22 counted every accepted move in forty runs per cell ($\lambda = 1.05, 1.10, 1.25$; $N = 64, 96$; $g = 1.5$). All 240 first exits were observed, and their means are consistent with Eq. (2) within 95% intervals that assume an exponential wait [1]. (At $N = 96$, $\lambda = 1.05$ the mean over all forty is 9302 sweeps; over the 39 that went on to convert, 7485.) Its registered verdict at $\lambda = 1.05$ was FALL-BACKS: exits return to the tube, and the first exit is not late. Our predicted verdict held and our predicted numbers did not: we expected 70 to 87% of exits to go through at $\lambda = 1.05$ and nearly all at 1.25. The author had given no prediction. For the 239 runs that reach a quarter conversion, the reciprocal of the mean number of exits per run is 0.56 to 0.68, with no trend in $\lambda$. This is a diagnostic, not a derived correction factor for the exit rate. Returns are a feature of the chain at every $\lambda$ tested, not a near-$\lambda = 1$ effect.

T59 counted fresh first exits by the move at the two ends of the window. At $\lambda = 1.05$ ($\tau_{AB} = 8329.7$ sweeps), 2000 runs at $N = 64$ and 1000 at $N = 96$ have means $0.992 \pm 0.023$ and $0.946 \pm 0.031$ of $\tau_{AB}$, with one first exit beyond $8\tau_{AB}$ where 1.0 is expected: ON THE COUNT at both sizes. At $\lambda = 1.30$, $N = 64$ ($\tau_{AB} = 419.0$), 4000 runs have mean $1.005 \pm 0.016$. Read by a look every five sweeps, the same runs give a mean later by $0.096 \pm 0.007\, \tau_{AB}$, and 94% of that comes from first exits that return before the look, about one first exit in thirteen: HIDDEN EXITS. The author's two predictions (chance; exits hidden between two looks) held, as did ours. Described, not scored: a mean wait read by five-sweep looks sits about 8 to 10% above the count for this reason alone, and the look leaves the wait's coefficient of variation (CV) unchanged (0.993 against 0.990 by the move). The same registration replayed, from their seeds, the long waits recorded earlier at $\lambda = 1.05$. A detected wait of 78,205 sweeps read as the tube at every look (TRUE WAIT). Seven long first exits were offered moves A and B at 2.94 to 3.03 and 1.94 to 2.04 per sweep throughout (OFFERED AS COUNTED). We read those long waits as chance.

*Exact, behind that reading.* At the sizes enumerated ($N = 48$ to 288, at $\lambda = 1.25$) the graphs a waiting tube reaches without changing its squares form three classes, each with the same count of moves at every price [1]. The chance of leaving is then the same at every attempt, so the first exit is exactly memoryless, with mean 8326 and

TABLE I. The tube's price list against $\lambda$ (exact). Wall: the cheaper of moves A and B. $\tau_{AB}$: Eq. (2) at $g = 1.5$, in sweeps. $\varepsilon$: the gap per vertex. Column: the cost of one curled column, $24\lambda - 16$, as a share of the gap at $N = 64$.

| $\lambda$ | wall | $\tau_{AB}$ | $\varepsilon$ | column |
|---|---|---|---|---|
| 1.05 | 15.2 | 8330 | 0.2 | 72% |
| 1.10 | 14.4 | 4844 | 0.4 | 41% |
| 1.15 | 13.6 | 2788 | 0.6 | 30% |
| 1.20 | 12.8 | 1570 | 0.8 | 25% |
| 1.25 | 12.0 | 845 | 1.0 | 22% |
| 1.30 | 11.2 | 419 | 1.2 | 20% |
| 1.35 | 10.0 | 183 | 1.4 | 18% |
| 1.40 | 8.0 | 68 | 1.6 | 17% |
| 1.45 | 6.0 | 22 | 1.8 | 16% |

417.5 sweeps at $\lambda = 1.05$ and 1.30, against $\tau_{AB}$'s 8329.7 and 419.0. Our arithmetic, unreviewed.

*Interpretation (ours, unverified).* Across the window, what $\lambda$ changes about the first exit is the two exponents of Eq. (2), and nothing else we have measured.

## IV. THE CONVERSION ACROSS THE WINDOW

*Protocol and the registered bar.* The tube at $g = 1.5$, $N = 64, 96, 144, 192$, is followed in blocks of five sweeps until 98% converted or 100,000 sweeps, and the final graph is saved. Let $d(v)$ be the number of pairs of edges at $v$ that close no square: 1 on the tube (curled-like), 2 on the flat torus (flat-like). It is a square count, not a dimension. Three criteria are registered per cell $(\lambda, N)$: (a) the CV of the wait lies in a stated band around 1, the value for a memoryless wait; (b) at least 80% of vertices have the square-count signature $d \in \{1, 2\}$ at half conversion; (c) the largest connected piece of flat-like vertices holds at least 70% of them at half conversion. A cell meeting all three is called sharp. The window verdict asks for a sharp change at every $\lambda$ from 1.05 to 1.30, at every size that passes its gates; $\lambda = 1.35$ is scored separately, as the edge. These are counting and connectivity statistics. They do not certify two geometric orders, a unique seed or a single moving front.

*Where the tube persists.* A tube persists in a cell when more than half its runs are under a quarter converted at sweep 200. In T8 that holds at every size from $\lambda = 1.05$ to 1.35 and fails at 1.40 and 1.45; our registered prediction for this edge, between 1.3 and 1.4, held, and the author had made none. (In T23, $\lambda = 1.35$ sat on the line at $N = 64$ and 96.) This edge is set by the stated criterion, not by where the barrier vanishes: the walls there are still 8 to 10 units, and move B costs energy up to $\lambda = 1.6$. Equation (2) gives a mean of 200 sweeps near $\lambda = 1.345$ but does not derive the edge [1].

*Three maps, each inconclusive.* T8 (thirty runs per cell, $\lambda = 1.05$ to 1.45, its $\lambda = 1.25$ cells taken from an earlier registered run) and T23 and T24 (120 runs per cell, $\lambda = 1.05$ to 1.35, fresh seeds) each returned INCONCLUSIVE by the letter of its own rules, for a different reason. T8's CV band was too tight for thirty waits and failed in nine of 24 persistent cells, and its endpoint-energy criterion failed at $\lambda = 1.35$. T23 sized the band to the sample; its energy criterion, which compared a window average with a saved endpoint, then failed at every size at $\lambda = 1.30$. T24 replaced that energy criterion with a check that each saved graph is valid, which held in all 28 cells; its CV criterion failed in four, $(N, \lambda) = (64, 1.25)$, $(64, 1.30)$, $(192, 1.30)$ and $(144, 1.35)$. In all three maps criteria (b) and (c) held in every cell through $\lambda = 1.30$. The author's prediction for T23 and T24, SHARP ACROSS THE WINDOW, was neither confirmed nor refuted by them.

*The detector.* The wait in those maps was a detector's. It rests for 200 sweeps, sets a threshold from the run's own spread of $S/N$ in that window, and then looks every five sweeps. A retrospective, exploratory reading of the saved rows (no new run, no verdict changed) shows that the detector's clock, not the chain, produced most of the long waits. The two waits that failed two of T24's cells, 24 and 76 times $\tau_{AB}$ at $N = 64$, were not waits: both tubes were three quarters or more converted when the window ended, and then rested on defective sheets until the lowered threshold was crossed. A registered tail test (T38; 10,000 runs at $\lambda = 1.25$ and 1.30) had returned TWO POPULATIONS, as the author's inferred prediction and ours had it; that verdict stands as scored. Read the same way, all 190 of its runs with no detector crossing had converted inside the window, and most of its fastest detected waits were exits inside the window. A registered replay of one T38 cell from its seeds (T58) returned every saved row unchanged. Its verdict on the three long waits left was MIXED, and neither registered prediction for them held (the author's, a hidden second curl; ours, a lowered threshold in all three). Timed with no detector, that cell's 4000 first exits have mean $1.071 \pm 0.017\,\tau_{AB}$ by five-sweep looks, with one beyond $10\tau_{AB}$ where 0.18 is expected. One long wait remains, of a length chance allows about one time in six [1].

*The fourth map (measured).* T60 repeats T24's grid and protocol with fresh seeds and one repair, written before the run. The wait is the first look, counted from sweep 0, at which a run no longer reads as the tube ($S = 5N/4$, $X = N$, every vertex at $d = 1$), with no resting window and no threshold, taken over every run of a cell. The bar is T23's and T24's, unchanged. Stage 1 ($\lambda = 1.05$ to 1.35; $N = 64$ to 192; 120 runs per cell) returned SHARP ACROSS THE WINDOW and, for the edge, BREAK-UP BEGINS AT THE EDGE. The author's standing prediction held on both, and ours. This is the fourth attempt at the window verdict and the first to return one, and the three inconclusive maps stand beside it as scored; in the registration's words, "A reader is entitled to weigh a fourth attempt less than a first." A red-team review of this verdict by a fresh reviewer is pending.

All three criteria hold in all 24 window cells, and (a) in all 28, including T24's three window failures (Table II). Described, not scored: the largest component's share falls with size at the top of the window, 0.93, 0.92, 0.87 and 0.83 at $N = 64, 96, 144, 192$ for $\lambda = 1.30$. At $\lambda = 1.35$ the cells at $N = 64$, 96 and 144 are sharp and $N = 192$ is not: its largest component holds 0.68, under the 0.70 line (0.67 in T23). The mean wait read by the look is $1.06\,\tau_{AB}$ over the window cells, the lag of Sec. III. T8's detected waits at $\lambda = 1.05$ had run 15 to 43% above the count; in the fourth map, timed with no detector, the cells at $\lambda = 1.05$ read 0.93 to 1.05 of it.

Whether the change stays sharp as the tube lengthens is scored at a held-out size, $N = 288$, alone, against numbers written from the four smaller sizes before that size ran (the registration's amendment of 10 October). *Mea-*

TABLE II. The fourth conversion map (T60, stage 1; measured) at $g = 1.5$. Each entry is the range of the four cell values at $N = 64, 96, 144, 192$, 120 runs per cell. CV: coefficient of variation of the wait, the wait being the first five-sweep look at which a run no longer reads as the tube (registered band 0.758 to 1.379). Signatures: mean share of vertices at $d \in \{1, 2\}$ at half conversion (registered line 0.80). Largest component: mean share of flat-like vertices in their largest connected piece at half conversion (line 0.70). Wait: mean wait over $\tau_{AB}$, read by the look; not scored. Zero energy: runs of 120 whose saved final graph has $H = 0$; not scored.

| $\lambda$ | CV of the wait | signatures | largest component | wait/$\tau_{AB}$ | zero energy |
|---|---|---|---|---|---|
| 1.05 | 0.936–1.006 | 0.999–1.000 | 0.99–1.00 | 0.93–1.05 | 117–120 |
| 1.10 | 0.868–1.034 | 0.999–1.000 | 0.98–1.00 | 0.96–1.19 | 118–120 |
| 1.15 | 0.897–1.095 | 0.998–0.999 | 0.97–0.99 | 0.91–1.16 | 113–118 |
| 1.20 | 0.932–1.119 | 0.996–0.998 | 0.95–0.98 | 1.01–1.18 | 110–115 |
| 1.25 | 0.846–1.004 | 0.990–0.994 | 0.90–0.98 | 1.01–1.20 | 98–109 |
| 1.30 | 0.928–1.090 | 0.974–0.986 | 0.83–0.93 | 0.92–1.13 | 82–94 |
| 1.35 (the edge) | 0.858–1.124 | 0.947–0.962 | 0.68–0.86 | 1.05–1.10 | 44–83 |

*sured:* AS PREDICTED AT THE HELD-OUT SIZE, and sharp at $N = 288$ at every $\lambda$ from 1.05 to 1.30. Every cell has its CV in the band (0.943 to 1.072) and each of its three numbers inside the range written for it. The largest component's share is 0.969, 0.959, 0.948, 0.911, 0.813 and 0.751 for $\lambda = 1.05$ to 1.30, against predicted values of 0.980, 0.965, 0.949, 0.942, 0.850 and 0.754, each with a range of 0.10 either side. The author's prediction for this size, recorded before it was read, matched ours: criteria (a) and (b) hold, and (c) is the one at risk at the top of the window. It held. *Interpretation (ours, not scored):* the straight line that predicted the held-out size also says where criterion (c) ends. At $\lambda = 1.30$ the largest component's share falls by about 0.08 for every hundred vertices and reaches the 0.70 line near 350, so this sharpness at the top of the window belongs to short tubes; tubes of thousands of vertices open from several places [2]. At the edge, $\lambda = 1.35$, the $N = 288$ cell is not sharp (0.557).

*The edge.* From $\lambda = 1.30$ to 1.35, pooled over sizes, the share of runs ending on a zero-energy graph falls from 0.73 to 0.50, and the share whose flat-like vertices are in more than one piece at a quarter conversion rises from 0.40 to 0.69 (T60). T24 gave 0.74 to 0.49 and 0.43 to 0.68; T23 gave 0.71 to 0.47, by its energy-window criterion, and 0.44 to 0.69. All three returned BREAK-UP BEGINS AT THE EDGE, which the author had predicted before T23, after seeing T8's shares. These counts show more disconnected flat-like regions near the persistence edge. They do not establish a single front at lower $\lambda$: several early pieces are common inside the window too.

## V. THE PRODUCT

*A retrospective reading of saved final graphs: exploratory; each graph's width is exact* [13]. A run that ends at zero energy ends on a flat torus: every edge on exactly two squares. That is a count at each edge, and a torus six edges around already satisfies it, so the energy does not say how wide the product is. The *width* of a flat torus is the length of its shortest noncontractible loop. It can be read exactly from the number of vertices within $r$ steps of a vertex, which equals the endless flat sheet's $2r^2 + 2r + 1$ until $2r$ reaches the width. The tube's width is 4. A flat torus needs at least 6: loops are even on a bipartite graph, and a noncontractible loop of four would itself be a square, a third on each of its edges.

Of the saved end states of the four maps ($N = 64$ to 288, $\lambda = 1.05$ to 1.45), 9,865 are exactly flat, and 9,858 of them have width 6: every one at $N = 144$, 192 and 288, and all but seven at 64 and 96 (six of width 8, one of width 12). Sampled against a graph built for the purpose, 41 of 41 at $N = 64$ and 96 are isomorphic to the tube re-glued two steps along: four vertices to a ring as before, but a walk around the ring arrives two steps along the tube, so no four steps close and the shortest way around is six. It has eight vertices at each distance, as the tube has.

So in these maps a clean birth is the birth of a flat torus six around, at every $\lambda$ and every size, and its width does not grow with $N$. Runs that kept a scrap can mix widths: about half hold a stretch wider than six, and in all 440 that hold both a six-around and a wider stretch a scrap lies between them [2]. The wall, the wait, the release and every registered verdict above are unaffected: they are energies and counts, blind to width. What is not shown, here or in Ref. [1], is a small direction becoming large. *Interpretation (ours, unverified):* every flat torus has zero energy, and at these couplings a flat torus accepts almost no move (leaving the one six around costs at least $16\lambda$, a wider one 32), so the chain stops at the first flat state it reaches, and the nearest to the tube is the tube re-glued two along. Where released energy stays in place the product can be wider: in a sealed control with one store per vertex (T48, $\lambda = 1.25$), 6 of the 15 tubes that converted are wider by the same kind of reading, and 9 are not. Heat kept in the stores instead does not widen it (T61 [1]); a price on loops of six does, with flaws throughout (T62 and T64, registered, read at 96 and 192

vertices; the held-out size is not read). What the rule charges for decides how wide the product is.

## VI. THE ROOM

Sealed, the released energy has only a reservoir to go to: $C$ stores of nonnegative energy, each move paid from or into one store chosen uniformly, so that graph plus stores conserve energy [1]. We call $C$ the room. It is a number of stores, not a heat capacity; conservation alone does not establish equilibrium. A conversion is called a birth here when its end state meets the sheet criterion of Ref. [1], a count ($f \geq 0.9$ with $f = \overline{d} - 1$, at least 90% of vertices at $d = 2$, under 5% at $d = 0$ or $d \geq 3$), not a topology test.

*Measured at* $\lambda = 1.25$ (T9, registered verdict BONFIRE WITH A THRESHOLD, as we had predicted; $N = 64, 96, 192$): for $C \leq N/16$ the converted region melts into the random phase in 17 to 20 of 20 runs, and for $C \geq N/2$ the runs are births in 18 to 20 of 20, the crossover lying between $N/4$ and $N/2$ at each size [1].

*Measured against* $\lambda$ (T18). One seed is planted in a sealed $24 \times 4$ tube ($N = 96$) with every store empty, at ten values of $C$ from $N/32$ to $2N$, twenty runs each, at $\lambda = 1.10$, 1.25 and 1.40. The smallest room giving a majority of births is $C^* = 12$, 36 and 72 stores. The registered reading is the ratio $R = C^*(1.40)/C^*(1.10) = 6$: PROPORTIONAL (the rule was $R \geq 2.5$). Our prediction held on the rule and undershot the size, since we expected $R \approx 4$; the author had given no prediction. Per unit of release, $C^*/(\varepsilon N) = 0.31$, 0.375 and 0.47, but the grid of $C$ steps by factors of 4/3 to 2 and cannot tell faster than proportional from proportional. What the data show is a room that grows with $\lambda$ at least as fast as the release. At $\lambda = 1.40$ even the largest reservoir, $C = 2N$, gave a birth in only 16 of 20 runs. This is one size. The melting coupling of the new sheet at $\lambda = 1.10$ and 1.40 was assumed close to that at 1.25, not measured; that it falls as $\lambda$ rises is our guess from $R = 6$, unverified.

*Melting moves with size (unfinished).* Two things on the record limit how far a statement at a fixed warm coupling carries. First, a retrospective reading of saved wiring (T37; $\lambda = 1.25$; tubes long enough to start in several places; its registered verdict, MANY SEEDS, MANY SCRAPS, stands by the letter): at $g = 1.5$, tubes of $4\times64$ to $4\times256$ vertices end with 94 to 97% of vertices flat-like, while at $4\times1024$ only 28% are flat-like, 55% have opened past flat ($d > 2$, the direction of the random phase), and the graphs took energy from the bath. At $g = 1.25$ the same length ends 95% flat-like. Whether the long warm runs had equilibrated is not known. Second, an early exploratory scan (one replica, under a variant of the energy that forbids surplus squares) put the coupling at which $S/N = 0.5$ at $g = 7.64$, 5.51, 4.34 and 3.51 for $N = 100$, 196, 400 and 900; repeating it at $\lambda = 1$ with several replicas and errors is not finished. Both point to a larger network melting at a lower coupling. A published entropy argument for a related graph model, in which the disordered phase's entropy grows as $N \ln N$ [10, 11], points the same way; applying it to this energy is ours, unverified, and it is a target to test against, not support. The maps of Sec. IV are at $N \leq 192$ and $g = 1.5$, and carry this caveat.

## VII. THE SCRAP ALONG THE WINDOW

At fixed coupling the share of runs that do not end on a zero-energy graph rises with $\lambda$ (Table II). In T8 every run at $\lambda = 1.05$ ended at zero energy, and all 120 saved endpoints are certified tori [1]; near the edge about half do not. The endpoints that hold energy rest on one curled column or on combinations of small defects.

After cooling, the scrap has been read only at $\lambda = 1.25$ [2]. In 1024-vertex sheets that opened from several starting places, the columns heal as the cooling slows, falling from 2.1 to 0.1 per sheet between cooling times of 10,000 and 300,000 size-fair sweeps (T51, registered verdict CLIFF; the author's prediction and ours, a gentle fall, both failed). The counts of two other kinds of piece do not change over that range. Read exactly from their wiring (T55, registered verdict KNOT), those pieces are stuck arrangements of their own, so the scrap is three kinds of knot of fixed energy: the column (14 units in its common form), an eight-vertex knot of two kinds of vertex (20 to 30) and a double column (8). The author's prediction, a seam (a piece with an edge joining distant places), failed; a scrap between stretches of different widths (Sec. V) is a seam of another sort [2]. The exact share of Sec. II is the single column's. T51 and T55 are at one $\lambda$; how the three kinds divide the scrap at other $\lambda$ has not been read.

## VIII. THE SAME PRICE LIST WITH MORE DIRECTIONS

In the author's hypothesis the space directions are tied: they curl and open together. That makes the number of directions a second axis of the map. With $2D$ links per vertex, $D = 3$ or 4, the same price list (16 per missing square, $4\lambda$ per surplus square, surplus counted above $2D - 2$ squares on an edge) gives a ladder that stays additive: each curled direction costs $\varepsilon$ per vertex. Every rung's cheapest single move was priced by listing switches (Table III).

*Exact.* (i) Each rung has its own wall, different at every $\lambda > 1$: nothing in this energy ties the directions together. (ii) The ratio of flat space's wall to the last rung's wall at $\lambda = 1.25$ is 2.7, 1.8 and 1.6 with two, three and four directions. Read as a margin between a bath that pays the last rung and one that damages flat space (our reading), more directions leave less, not more; a rule of ours built on this ratio, a window wherever it is at least 2, failed in T39. (iii) A state curled in every direction at

TABLE III. More directions. Exact: the wall (the cheapest single move out of each ideal curled state), the range of $\lambda$ over which it is positive, and its value at $\lambda = 1.25$. Measured: registered verdicts in capitals; sealed unless stated, named vertices, no tie between directions. Wide flat space's own wall is 32, 64 and 128 with four, six and eight links, at every $\lambda$ (the four-link torus six around has a lower one, $16\lambda$). On the $4 \times 4 \times 4 \times 12$ torus of T48 the cheapest move at $\lambda = 1.40$ is a different one, $192 - 136\lambda = 1.6$. Six- and eight-link rows carry the reproduction caveat of the text. Whose predictions held is in the text and the record [13].

| links | curled directions | wall | wall $> 0$ for | at 1.25 | registered runs |
|---|---|---|---|---|---|
| 4 | one: the tube, $4 \times L$ | $32 - 16\lambda$, then $64 - 40\lambda$ | $1 < \lambda < 1.6$ | 12 | Secs. III to VII |
| 4 | both: a gas of 4-cubes | $32 - 32\lambda$ | never | downhill | none; no wall at any $\lambda > 1$ (exact move listing) |
| 6 | one: $4 \times L \times L'$ | $96 - 48\lambda$ | $1 < \lambda < 2$ | 36 | T41 NEVER OPENS at $\lambda = 1.25$ and 1.40, reservoirs $N/8$ to $4N$ (0 of 144 runs); T48 STAYS (32 of 32) |
| 6 | two: $4 \times 4 \times L$, $L \geq 18$ | $96 - 64\lambda$, then $128 - 88\lambda$ | $1 < \lambda < 1.45$ | 16 | T32 FIXED WALL (16 at 192 to 512 vertices); T30 NEVER OPENS; T39 NO WINDOW; T48 ADVANCES ONLY at 1.25, DAMAGED at 1.40; T53 (warm bath) ADVANCES ONLY |
| 6 | all three: a gas of 6-cubes | $96 - 80\lambda$ | $1 < \lambda < 1.2$ | downhill | T30-gas FIRST ONLY at 1.10; T49 (one cube) PART OPEN at 1.10 to 1.25 |
| 8 | one | $160 - 64\lambda$ | $1 < \lambda < 2.5$ | 80 | not run |
| 8 | two: $4 \times 4 \times 8 \times 8$ | $160 - 96\lambda$ | $1 < \lambda < 1.67$ | 40 | T39 NO WINDOW at 1.25 and 1.50 (every cell STUCK) |
| 8 | three: $4 \times 4 \times 4 \times L$ | $160 - 112\lambda$ | $1 < \lambda < 1.43$ | 20 | T33 NO CASCADE (every run STALLS); T40 NEVER STARTS (pushes 20 to 160); T48 STAYS at 1.25, ADVANCES ONLY at 1.40 |
| 8 | all four: a gas of 8-cubes | $160 - 128\lambda$ | $1 < \lambda < 1.25$ | 0 | T33 NO CASCADE and T40 NEVER STARTS at 1.10; T49 (one cube) STUCK at 1.10 and 1.15, PART OPEN at 1.30 |

every vertex is connected only up to $4^D$ vertices, where it is the hypercube, so a fully curled state larger than one cube is a gas of separate pieces. This last applies a published theorem [12] whose proof we have not read.

*Measured.* By the registered verdicts, which rest on counts of flat-like vertices and their largest connected piece, no setting with six or eight links opened a curled state into one flat space. The price of the first move out of the two-curled six-link torus is fixed with size (T32; the author's prediction and ours held). Given that push at $\lambda = 1.25$, its first curled direction opened fully in 28 of 84 runs and its second never in a cold reservoir (T30; both predictions failed). The one-curled six-link torus, the direct counterpart of Sec. VI, never opened at any reservoir from $N/8$ to $4N$ (T41), so the room it would need could not be measured. With eight links a push equal to the first move stalled after that move (T33), and pushes up to eight times it moved no run off its starting rung by the registered measure (T40). The author's predictions, some of them inferred from her stated positions, failed in T30, T39 and T41 and were not reached in T33 and T40, where nothing opened. Ours were right that nothing opens all and wrong in particulars: the windows we expected at larger $\lambda$ did not appear (T39), and our estimate of the push that starts four directions was too low.

Three results cut the other way. In a gas of 6-cubes at $\lambda = 1.10$, two of three directions opened in 21 of 24 runs, in many small flat-like patches and never one space (T30-gas). With released energy kept at the vertex that releases it (T48), the two-curled torus at $\lambda = 1.25$ advanced one rung and stopped, as the author and we had predicted (ADVANCES ONLY). At $\lambda = 1.40$ that prediction failed: the setting's registered verdict is DAMAGED, with damage in 7 of 16 runs, and in 5 of the 16 the torus reached a connected flat-like region holding at least half its vertices. And in a warm bath at fixed coupling the same torus returned ADVANCES ONLY at every length and both $\lambda$ (T53): its first direction opens and it rests there. Its second opened only in a minority of runs, at the two shorter lengths and $\lambda = 1.40$, in baths that then damaged the space. The author's prediction of a window failed, and her prediction of several starting places held. Ours held at $\lambda = 1.25$ and failed at 1.40, where we had expected a window at the shortest length.

The verdict words OPENS and ONE SPACE are counts too. By the distance profiles of their saved graphs, about a third of the six-link runs that met the count's criterion for an opening are the starting torus glued together differently, not a wider space [3].

*Interpretation (ours, unverified).* The bath that crosses the last wall is close to the bath that takes flat

space apart, and a single move does not price the barrier of a growing patch in three directions.

*With a tie.* A term that ties the directions is an ingredient the author has added, a further step from CQG. The tied runs read so far returned NOT ALL (T44 to T46), as we had predicted in outline; the author's prediction, that one push opens all, failed. Under the tie the author chose afterwards, in which nothing is released until a vertex's last direction opens, the single-move wall of the one-curled torus falls from 39.2 at $\lambda = 1.05$ to 4.0 at 1.25 while flat space's stays 64 (exact). Its first registered test held that torus and the two-curled one in a warm bath (T56). No cell has an OPENS majority: the one-curled torus STAYS at $\lambda = 1.10$ to 1.20 and ADVANCES ONLY at 1.25, the two-curled one STAYS, and an untied control shows NO DIFFERENCE. The author's prediction, that it opens into one flat space, failed, and so did ours, a window at 1.15 and 1.20. The tie as written charges a damaged vertex nothing, and the tied torus broke where the untied one was untouched. The test that prices the barrier of a growing patch has no verdict yet. Reference [3] treats both.

*The caveat on every six- and eight-link statement.* These are statements about an energy we derived from the published definitions and a kernel validated on its own terms (at four links it reproduces the four-link chain draw for draw, and its energies and moves are checked by brute force). The published six-link curve [9] has not been reproduced: the gate failed and stays open. A factor of two in the coupling is accounted for by the published equation; whether the remaining difference is the published figure's protocol or its graph ensemble is not settled. There is no published eight-link curve.

## IX. DISCUSSION

*The trade-off.* In two directions the map has one shape. As $\lambda$ rises from 1.05 to 1.35 at $g = 1.5$, the gap per vertex, the most a conversion can release, grows linearly (0.2 to 1.4), the wall falls (15.2 to 10), the counted wait falls from 8330 to 183 sweeps, the largest flat-like component at half conversion shrinks, fewer runs end at zero energy, and a sealed conversion needs at least proportionally more room. As $\lambda$ falls toward 1 everything the change produces shrinks to nothing and the tube lasts longer. No setting at $g = 1.5$ gives a large release, a long-lived tube and a product that needs little room.

*Which settings give a birth.* Within what was run: in a bath at $g = 1.5$, the tube at $\lambda = 1.05$ to 1.30 and 64 to 288 vertices converts with the fourth map's count criteria met in every cell, and 72 to 120 of 120 runs per cell end on a zero-energy graph, a flat torus six edges around (Sec. V). Sealed, at 96 vertices, a majority of runs are births when the room is at least 12, 36 and 72 stores at $\lambda = 1.10$, 1.25 and 1.40. Outside that: breakup from $\lambda = 1.35$ and no persistence from 1.40; in long warm tubes, melting; with six and eight links, no setting with a registered verdict of one flat space.

*The author's framing.* In the author's hypothesis a selection among settings would act on this trade-off. The toy maps the ingredients; it cannot count what a cycle yields.

*What this cannot show.* Anything at exactly $\lambda = 1$, where nothing is released. Anything in the maps at couplings other than $g = 1.5$, or at sizes above 288. The room at any size but 96. Anything with interchangeable vertices: all runs treat vertices as named, and weighting each unlabeled class once is a different ensemble, with acceptance $\min\{1, [A(G')/A(G)]\, e^{-\Delta H/g}\}$ for symmetry counts $A$. It leaves every energy here unchanged and changes the rates; a first-move suppression by a factor $cN$ corresponds to an increment $g \ln(cN)$ in an effective barrier, not to a derived lifetime. And anything about reality beyond whether a mechanism is possible in a class of toys.

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[13] The record this paper relies on, in `https://github.com/EmilySmithCreate/RecreationalPhysics`. `PREREGISTRATION.md`: T7, T8, T9, T18, T22, T23, T24, T30, T32, T33, T37, T38, T39, T40, T41, T44 to T46, T48, T49, T51, T53, T55, T56, T58, T59, T60 and Gate C′. `ASSUMPTIONS.md`: B4, Q21, the first-look section, O14, O34, O38 to O44, O49, O50, O52 to O55, O62, O64, O65, O79, O81, O84, O85, O88, O90 to O92, O96, O100, O103 to O110, O112, O113 and O115 to O117. `VISION.md`: Updates 22 to 26, 30, 37, 43, 46, 47, 53 and 54. `TASKS.md`: T9. `docs/papers/measurement_scope.md` states what the square counts establish.